Vidyavardhini's College of Engineering & Technology

Department of Computer Science Engineering (Data Science)

Academic Year: 2026-27

EXPERIMENT ASSESSMENT

ACADEMIC YEAR 2026-27

Course: Wireless and Mobile Communication **Course code:**

Year: TE **SEM: V**

Experiment No. 6
Aim : To analyze channel interference in the 2.4 GHz Wi-Fi band using a Wi-Fi Analyzer application.
Name: Meet Mehta
Roll Number: 30
Date of Performance:
Date of Submission:

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission.	10	
Total	**20**	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	5	3	2
Understanding	5	3	2
Journal work and timely submission.	10	8	4

Checked by

Name of Faculty : Mrs. Sonali Sable

Signature : **Date** :

Aim: To analyze channel interference in the 2.4 GHz Wi-Fi band using a Wi-Fi Analyzer application.

Theory

1. Wi-Fi Channel Interference

Wi-Fi channel interference occurs when multiple wireless networks operate on the same or overlapping channels and their signals interfere with each other.

Interference can reduce the quality and performance of wireless communication.

2. 2.4 GHz Wi-Fi Band

The 2.4 GHz Wi-Fi band is widely used by wireless networks and contains multiple Wi-Fi channels.

Since many nearby Wi-Fi networks may operate in this band, some channels may become crowded and experience higher interference.

3. Channel Overlap

In the 2.4 GHz band, neighboring Wi-Fi channels can overlap with each other.

When two networks operate on overlapping channels, their transmitted signals can interfere, particularly when the networks have strong signals and are located close to each other.

4. Channel Occupancy

Channel occupancy represents the number of nearby Wi-Fi networks operating on a particular channel.

A channel with many networks has higher channel occupancy and may be more likely to experience interference.

For example, if several nearby networks are operating on the same channel, the channel is considered crowded.

5. RSSI and Interference

RSSI (Received Signal Strength Indicator) represents the received strength of a Wi-Fi signal and is expressed in dBm.

A nearby network with a stronger signal can have a greater impact on the wireless environment than a network with a very weak signal.

Therefore, channel occupancy, channel overlap, and signal strength should be considered together when analyzing possible interference.

6. Channel Analysis

A Wi-Fi Analyzer application provides graphical and numerical information about nearby Wi-Fi networks.

The Channel Graph can be used to observe:

- Wi-Fi networks operating on different channels
- Channel overlap
- Relative signal strengths
- Crowded channels
- Less-crowded channels

The Channel Rating view, if available in the application, can also help compare the suitability of different channels.

Procedure

Step 1: Open the Wi-Fi Analyzer Application

Install and open a suitable Wi-Fi Analyzer application on the smartphone.

Step 2: Enable Wi-Fi and Required Permissions

Enable Wi-Fi on the smartphone.

Provide the required permissions, such as Location permission, if requested by the application.

Step 3: Start Wi-Fi Scanning

Start the Wi-Fi scan to detect nearby wireless networks.

Allow the application sufficient time to display the detected networks.

Step 4: Select the 2.4 GHz Band

Select the 2.4 GHz frequency band for analysis, if the application provides a band-selection option.

Step 5: Observe Detected Networks

View the list of Wi-Fi networks operating in the 2.4 GHz band.

Observe the information provided by the application for each network.

Step 6: Record the SSID

Select or note the SSID (name) of each detected Wi-Fi network.

Step 7: Record the Channel Number

Observe and record the Wi-Fi channel number used by each detected network.

Step 8: Record RSSI

Observe the RSSI/signal strength of each detected network and record the value in dBm.

Step 9: Open Channel Analysis

Open the Channel Graph or Channel Rating view in the Wi-Fi Analyzer application.

Step 10: Observe Channel Occupancy

Observe the number of Wi-Fi networks operating on different 2.4 GHz channels.

Identify channels having a large number of detected networks.

Step 11: Identify Crowded and Less-Crowded Channels

Identify the most crowded and least-crowded channels based on the number of detected networks.

Step 12: Observe Channel Overlap

Observe the overlap between adjacent Wi-Fi channels in the Channel Graph.

Note the channels where multiple networks overlap.

Step 13: Compare Signal Strengths

Compare the RSSI values of networks operating on the same or overlapping channels.

Observe whether networks with stronger signals are present on crowded or overlapping channels.

Step 14: Identify Channels with Higher Interference Potential

Identify the channels that are likely to experience higher interference based on:

1. Channel occupancy
2. Channel overlap
3. Signal strength (RSSI)

Step 15: Record Observations and Conclusion

Record all observations in the observation tables.

Based on the channel occupancy, overlap, and RSSI values, write the conclusion regarding channel interference in the 2.4 GHz Wi-Fi band.

OUTPUT:

Observation Table:

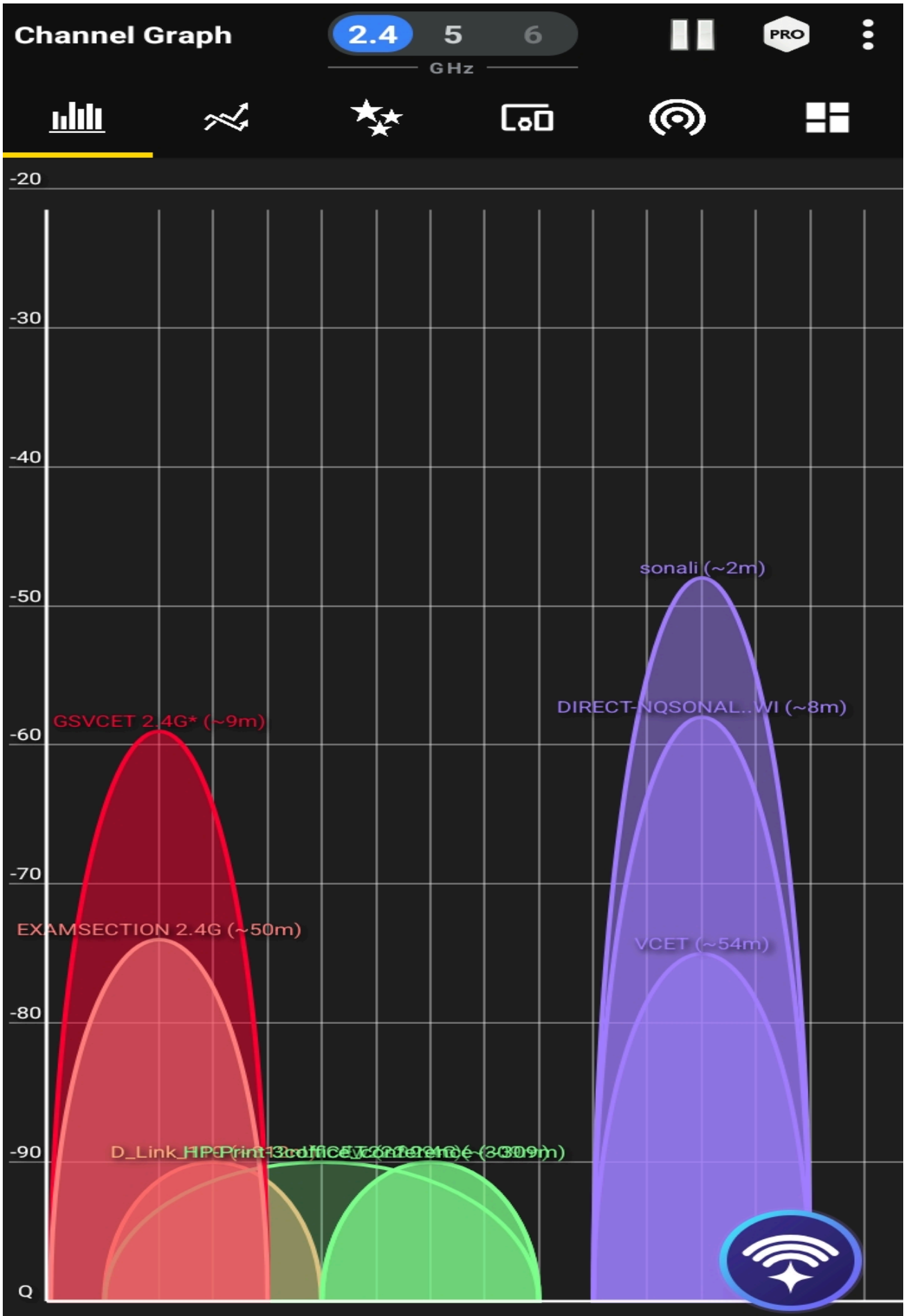
Sr. No.	SSID	RSSI (dBm)	Channel No.	Frequency Band	Interference Observation
1	Sonali	−53	11	2.4 GHz	Strong signal; crowded channel
2	DIRECT-NQSONALIJGWI	-55	11	2.4 GHz	Strong signal; crowded channel
3	GSVCET 2.4G	-56	1	2.4 GHz	Strong signal; channel has multiple networks
4	VCET	-68	11	2.4 GHz	Moderate signal; crowded channel
5	EXAMSECTION 2.5G	-73	1	2.4 GHz	Moderate signal; channel has multiple networks
6	office_conference	-85	6	2.4 GHz	Weak signal; less interference potential
7	HP-Print-3c-LaserJ	-90	6(4)	2.4 GHz	Weak signal; channel overlap present

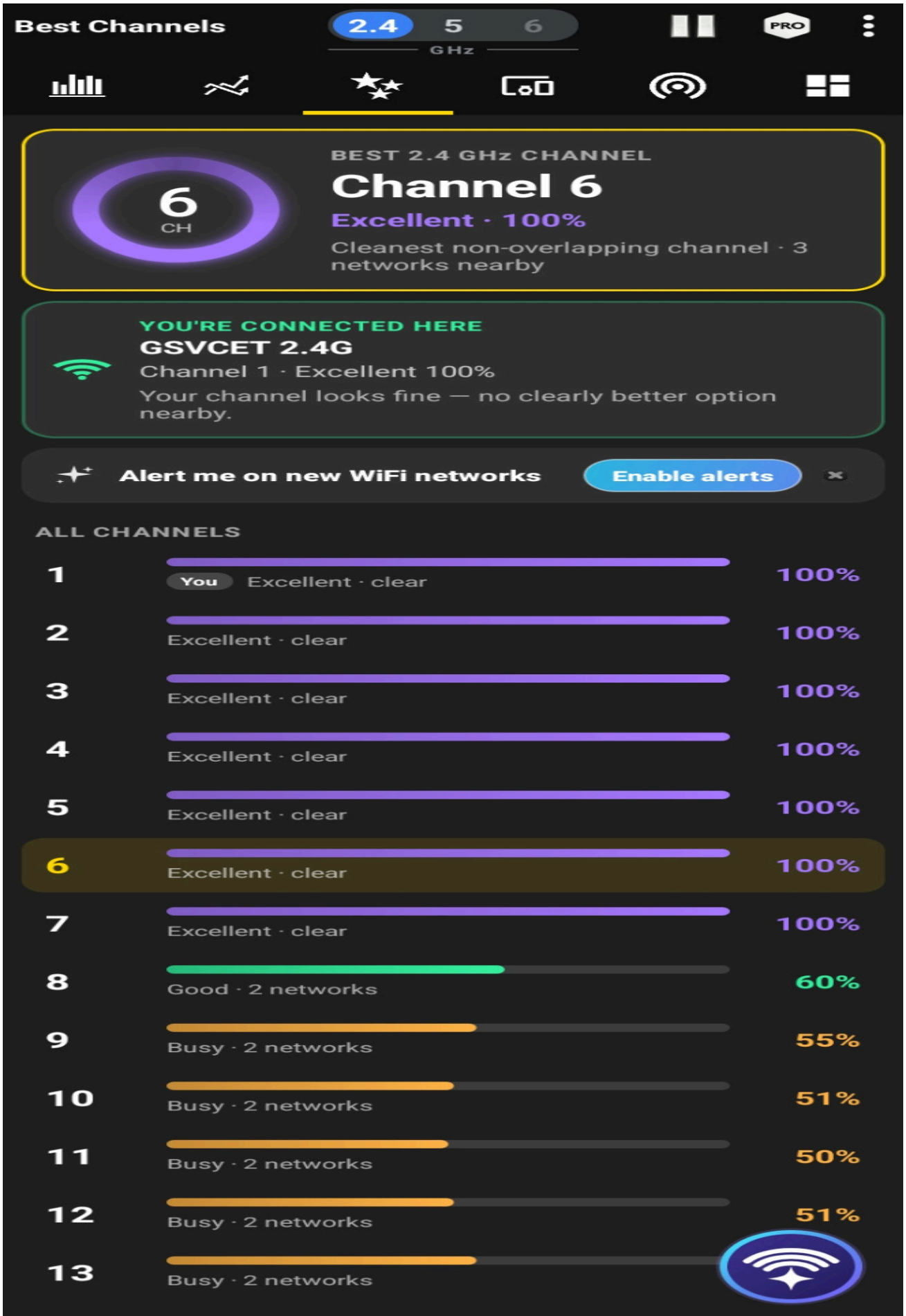
Sr. No.	SSID	RSSI (dBm)	Channel No.	Frequency Band	Interference Observation
	et Pro..dw				
8	D_Link_ 110	-90	2	2.4 GHz	Weak signal; low interference potential

Note: Record the observations for each detected Wi-Fi network individually in the observation table. The entries should match the information displayed by the Wi-Fi Analyzer application.


Access Points
2.4
5
6
GHz
PRO
IN RANGE 8
by Strength
sonali
WiFi 4
aa:84:9e:6a:c7:c4 4m
2462MHz (20MHz) CH 11
WPA2
-53
dBm
DIRECT-NQSONALIJGWI
WiFi 4
d6:bb:43:1f:3d:cb 5m
2462MHz (20MHz) CH 11
WPA2
-55
dBm
GSVCET 2.4G
WiFi 4
64:29:43:b7:0b:c0 6m
2412MHz (20MHz) CH 1
WPA2 26Mbps [D-Link]
-56
dBm
VCET
WiFi 4
e8:48:b8:3d:3b:02 24m
2462MHz (20MHz) CH 11
WPA2 [TP-Link]
-68
dBm
EXAMSECTION 2.4G
WiFi 4
64:29:43:b7:0b:f0 44m
2412MHz (20MHz) CH 1
WPA2 [D-Link]
-73
dBm
office_conference
WiFi 4
10:27:f5:37:55:b2 174m
2437MHz (20MHz) CH 6
WPA2 [TP-Link]
-85
dBm
HP-Print-3c-LaserJet Pro..dw
WiFi 4
34:68:95:0e:a9:3c 390m
2437MHz (40MHz) CH 6(4)
WPA2 [Hon]
D_Link_110
WiFi 4
6c:19:8f:e1:6f:10 393m


Access Points
2.4
5
6
GHz
PRO
IN RANGE 9
by Strength
2462MHz (20MHz) CH 11
WPA2
-60
dBm
GSVCET 2.4G
WiFi 4
64:29:43:b7:0b:c0 12m
2412MHz (20MHz) CH 1
WPA2 26Mbps [D-Link]
-62
dBm
VCET
WiFi 4
e8:48:b8:3d:3b:02 27m
2462MHz (20MHz) CH 11
WPA2 [TP-Link]
-69
dBm
EXAMSECTION 2.4G
WiFi 4
64:29:43:b7:0b:f0 28m
2412MHz (20MHz) CH 1
WPA2 [D-Link]
-69
dBm
office_conference
WiFi 4
10:27:f5:37:55:b2 276m
2437MHz (20MHz) CH 6
WPA2 [TP-Link]
-89
dBm
D_Link_110
WiFi 4
6c:19:8f:e1:6f:10 312m
2417MHz (20MHz) CH 2
WPA2 [D-Link]
-90
dBm
HP-Print-3c-LaserJet Pro..dw
WiFi 4
34:68:95:0e:a9:3c 309m
2437MHz (40MHz) CH 6(4)
WPA2 [Hon]
-90
dBm
VCET222 2.4G
WiFi 6
dc:ea:e7:0c:5b:60 309m
2437MHz (20MHz) CH 6


Channel Graph
2.4
5
6
GHz
PRO
-20
-30
-40
-50
-60
-70
-80
-90
Q
sonali (~2m)
DIRECT-NQSONAL..WI (~8m)
GSVCET 2.4G* (~9m)
EXAMSECTION 2.4G (~50m)
VCET (~54m)
D_Link_HP-Print-13-office_conference (~309m)


Best Channels
2.4
5
6
GHz
PRO
BEST 2.4 GHz CHANNEL
Channel 6
6
CH
Excellent · 100%
Cleanest non-overlapping channel · 3 networks nearby
YOU'RE CONNECTED HERE
GSVCET 2.4G
Channel 1 · Excellent 100%
Your channel looks fine — no clearly better option nearby.
Alert me on new WiFi networks
Enable alerts
ALL CHANNELS
1 You Excellent · clear 100%
2 Excellent · clear 100%
3 Excellent · clear 100%
4 Excellent · clear 100%
5 Excellent · clear 100%
6 Excellent · clear 100%
7 Excellent · clear 100%
8 Good · 2 networks 60%
9 Busy · 2 networks 55%
10 Busy · 2 networks 51%
11 Busy · 2 networks 50%
12 Busy · 2 networks 51%
13 Busy · 2 networks

Channel Analysis Observation

Channel No.	No. of Networks	Strongest RSSI (dBm)	Channel Occupancy	Interference Observation
1	**2**	**-56**	**100%**	**Multiple networks' moderate interference potential**
6	**3**	**-85**	**100%**	**Multiple networks; channel overlap/ interference potential**
11	**3**	**-53**	**50%**	**Strongest signal present; crowded channel**

Note: The values recorded in the observation table must match the values displayed in the corresponding screenshots attached to the experiment.

Conclusion: From the Wi-Fi Analyzer results, it was observed that channel occupancy, channel overlap and RSSI affect the interference level in the 2.4 GHz Wi-Fi band. Channels 1 and 11 had multiple nearby networks with relatively strong signals, indicating higher interference potential. Channel 6 was shown as the best channel by the analyzer, with fewer overlapping networks. Thus, Wi-Fi Analyzer can be used to identify crowded channels and select a comparatively suitable channel.

VIVA Questions

What is Wi-Fi channel interference?
It occurs when multiple Wi-Fi networks operate on the same or overlapping channels, causing their signals to interfere.

What is channel overlap?
Channel overlap occurs when neighboring Wi-Fi channels share part of their frequency range, causing possible interference.

What is channel occupancy?
Channel occupancy represents the number of nearby Wi-Fi networks operating on a particular channel.

Why are channels 1, 6, and 11 commonly used in the 2.4 GHz Wi-Fi band?
They are commonly used because they are non-overlapping channels in the 2.4 GHz band.

What factors are considered when identifying channels with higher interference potential?
Channel occupancy, channel overlap, and RSSI (signal strength).